

1. Develop a Python application that validates: Email Address Password Mobile Number

```
import re

email = input("Enter Email Address: ")

password = input("Enter Password: ")

mobile = input("Enter Mobile Number: ")

email_pattern = r'^[A-Za-z][A-Za-z0-9._]*@[A-Za-z]+\.(com|org|edu|net|in)$'

password_pattern = r'^(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[@#%&!])[A-Za-z\d@#%&!]{8,}$'

mobile_pattern = r'^[6-9]\d{9}$'

if re.fullmatch(email_pattern, email):

    print("Valid Email")

else:

    print("Invalid Email")

if re.fullmatch(password_pattern, password):

    print("Strong Password")

else:

    print("Weak Password")

if re.fullmatch(mobile_pattern, mobile):

    print("Valid Mobile Number")

else:

    print("Invalid Mobile Number")
```

2. DFA Simulator for Strings Ending with "ab"

```
transition = {

    'q0': {'a': 'q1', 'b': 'q0'},

    'q1': {'a': 'q1', 'b': 'q2'},

    'q2': {'a': 'q1', 'b': 'q0'}

}

initial_state = 'q0'
```

```

final_states = ['q2']
n = int(input("Enter number of input strings: "))
for i in range(n):
    string = input(f"\nEnter String {i+1}: ")
    current_state = initial_state
    path = [current_state]
    valid = True
    for symbol in string:
        if symbol not in ['a', 'b']:
            valid = False
            break
        current_state = transition[current_state][symbol]
        path.append(current_state)
    print("Transition Path:")
    print(" → ".join(path))
    if valid and current_state in final_states:
        print("Accepted")
    else:
        print("Rejected")

```

3. Develop a Python-based text search engine that performs pattern matching using Regular Expressions.

```

import re

text = """Meeting on 12/09/2026
Call 9876543210
#NLP
@OpenAI
natural language processing"""

```

while True:

```
print("\n----- TEXT SEARCH ENGINE -----")
```

```
print("1. Search Date")
```

```
print("2. Search Phone Number")
```

```
print("3. Search Hashtag")
```

```
print("4. Search Mention")
```

```
print("5. Search Prefix")
```

```
print("6. Search Suffix")
```

```
print("7. Exit")
```

```
choice = int(input("Enter your choice: "))
```

```
if choice == 1:
```

```
    result = re.findall(r'\b\d{2}/\d{2}/\d{4}\b', text)
```

```
    print("Dates Found:", result)
```

```
elif choice == 2:
```

```
    result = re.findall(r'\b[6-9]\d{9}\b', text)
```

```
    print("Phone Numbers Found:", result)
```

```
elif choice == 3:
```

```
    result = re.findall(r'#\w+', text)
```

```
    print("Hashtags Found:", result)
```

```
elif choice == 4:
```

```
    result = re.findall(r'@\w+', text)
```

```
    print("Mentions Found:", result)
```

```
elif choice == 5:
```

```
    prefix = input("Enter Prefix: ")
```

```
    pattern = r'\b' + re.escape(prefix) + r'\w*'
```

```
    result = re.findall(pattern, text, re.IGNORECASE)
```

```
    print("Matching Words:", result)
```

```
elif choice == 6:

    suffix = input("Enter Suffix: ")

    pattern = r'\b\w*' + re.escape(suffix) + r'\b'

    result = re.findall(pattern, text, re.IGNORECASE)

    print("Matching Words:", result)

elif choice == 7:

    print("Exiting Program...")

    break

else:

    print("Invalid Choice")
```